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Introduction

Hello, and welcome to this course what is Cloud Computing. This is a question that many people ask, because we hear the term all the time in the IT industry and beyond.

- But what is it?
- What does it comprise of?
- why is it so prevalent and key to business development?

This course will aim to answer all of those questions.

This course is intended for anyone who wants to learn about Cloud Computing, who may have no or very little knowledge of it currently. It's for those of you who are looking to learn more about the Cloud, to decide if it's something you want to adopt within your business, or those of you that may be seeking a career move and want to learn the foundation of Cloud principles, then this course is for you. As this is a foundation and beginner course at Cloud Computing, I shall be discussing explaining the following key areas. Starting with, what is Cloud Computing. Here I should give you a definition of what it is in its most simplest form. We'll then look at Cloud deployment models, and within this lecture, I shall be reviewing and discussing three different models that you should be aware of. Next, I'll look at some key Cloud concepts, and here I shall explain some of the key concepts that apply to Cloud Computing. Following this, I'll be looking at different Cloud service models, and these differ from deployment models and focus more on the services offered by the Cloud. So I will dive into each of these, giving you an explanation of how they differ. Then I'll look at some common use cases of Cloud Computing, and here I should give a few examples of what enterprises are using Cloud Computing for, to drive their business and operations forward.

Then I'll look at how data center architecture is reflected within the Cloud. So in this lecture, I explain how certain technology that we use on premises today, such as routes and switches, are reflected within the public Cloud. And then finally, there'll be a summary lecture, highlighting the key points from all the previous lectures. On completion of this course, you will have a comprehensive understanding of Cloud Computing along with its key concepts and benefits. This will allow you to determine when Cloud Computing should be used and which model and service to deploy. It will also give you a solid foundation to continue on with deeper learning about the Cloud and other Cloud service providers, such as Amazon Web Services, AWS, Microsoft Azure, and the Google Cloud Platform, GCP. Although not essential for this course, it would be advantageous if you have a basic understanding of Server hardware components and a basic understanding of what a datacenter is.

What is Cloud Computing

The phrase cloud computing has been used heavily within the IT industry for many years now, and more recently it's being used within other sectors such as retail and finance, as it becomes increasingly popular. Many people often refer to the technology simply as the Cloud. Cloud computing is a rapidly growing technology, and the adoption of this is a key strategy for many organizations, and there is a very good reason behind this. It's changing the landscape of how many companies operate on a huge scale, with significant business and technical advantages and benefits that can't be ignored. The growth of cloud computing has been exponential over the past few years, so what is it? Put simply, cloud computing is a remote virtual pool of on-demand shared resources offering to compute, storage, database and network services that can be rapidly deployed at scale. Now there may be a couple of terms within this definition that are new to you, or not too clear, such as virtual or computer. Don't worry, I'm going to break these down over the next couple of slides, after which I shall give you the definition again, and you should be able to fully understand this concept.

Before fully understanding cloud computing, we must be aware of some existing technology that it's based upon, that being, virtualization, and this is being used in on-premise data centers for a long time. But this virtualization maximizes the power of cloud computing, and without this virtualization, it would just not be possible. So what is virtualization? In essence, it allows the possibility of having multiple virtual machines, VM's, each running essentially a separate operating system and applications, all installed on one physical server. These VM's all run at the same time without being aware of each other's existence while sharing the underlying hardware resources. This sharing of hardware resources is a key element of understanding of virtualization and is achieved through a hypervisor.

A hypervisor is a piece of software you use to create the virtualized environment, allowing for multiple VM's to be installed on the same host. When installed, the hypervisor sits logically between the physical server hardware and the virtual machines and creates a shared pool of virtual hardware resources for each of them to access. All VM's installed on the host see the hardware as they normally would.

However, any request to the hardware goes via the hypervisor, which then handles that access ensuring the hardware resources are shared between all other VM's as needed and as configured. Now there are obvious benefits of virtualization, which include:

- **reduced capital expenditure (investicija)**. As less hardware is required, as you have the ability to provision multiple VM's on a single host.
- **Reduced operating cost (stroški obratovanja)**, because as there is less hardware, there is less space, power, cooling required within your data center, and
- **the footprint also reduces** within your data center as less space is required to house your server hardware.
- This **optimization of resources (optimizacija virov)** when in a cloud environment means everyone can benefit from virtualization, from the cloud vendor to the consumer.

Now just a quick note before we leave this topic of virtualization. A VM within the public cloud is sometimes referred to as an instance, this term is very vendor specific but it refers to the same object as a virtual machine.

When discussing **resources** within cloud computing, it won't be long before you come across the terms such as:

- compute,
- storage,
- database and
- network resources
- machine learning or
- artificial intelligence.

However, for this beginners course, I just want to keep the focus on the core foundation services.

It's a good idea to have a clear distinction between compute, storage, database, and network. And what each of these refers to, is that it will certainly help you going forward when identifying what services you want to move to the cloud, should you decide to do so. Compute objects provide the brains to process your workload. Including what's required to process a run request from applications and services. As a comparison, if you think of hardware devices with CPU's and RAM, typically servers and how they work in a classic, on-premises environment, compute resources in a cloud, are comparable to these. Storage resources simply allow you to save and store your data. Any resource that allows you to save your data in the cloud, is classed as a storage resource. Again as a comparison, in the typical environment these will be seen as your server hard disks, or your network-attached storage, which provides file-level shed storage over the network, or your high-speed storage area network, your SAN, which is block-level storage accessed over a high-speed network. Database resources allow you to store structured sets of data used by your applications. Again as a comparison, databases are widely used in Data Centers with some common database engine types being SQL Server, Oracle, and MySQL. Within the Cloud, there are a wide variety of database engines available for different use cases. Network resources provide the connectivity allowing all other resources, compute, storage, and database, to communicate with each other. In a typical environment, you would find hardware, such as routers to route traffic between

network switches, which provide the backbone of network connectivity allowing the host to talk to one another and firewalls to allow or deny traffic into the environment.

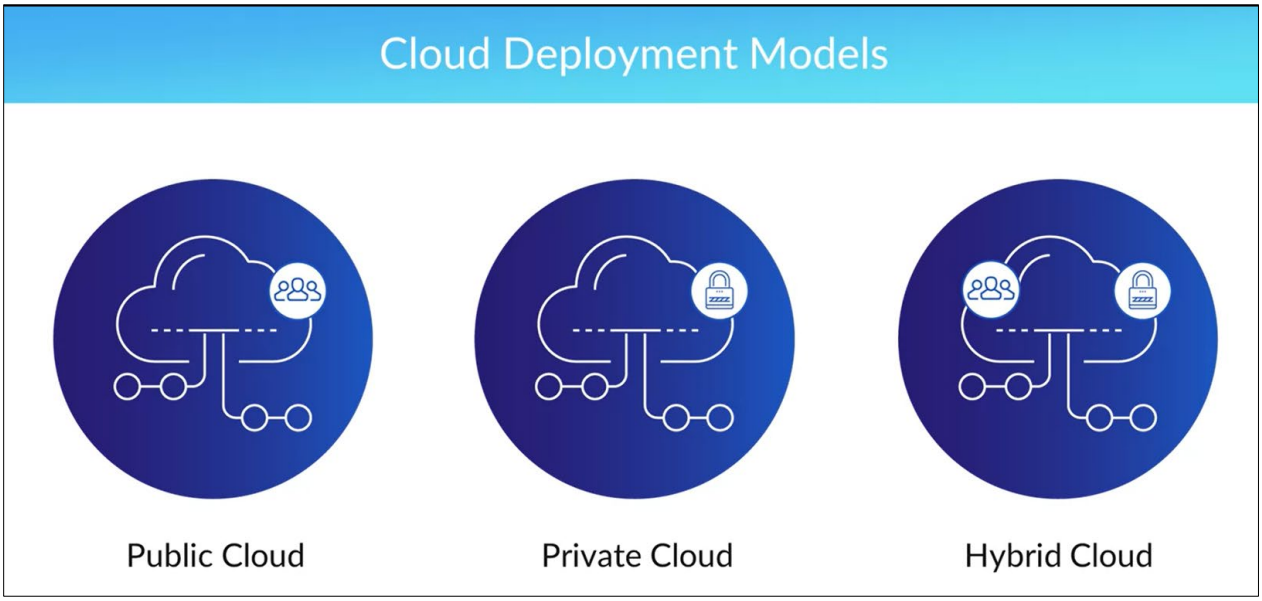
So now if we go back to our definition of cloud computing given earlier of, cloud computing is a remote virtual pool of on-demand shared resources, offering to compute, storage, database and network services that can be rapidly deployed at scale. You should now have a clearer understanding of what this actually means. The red section refers back to the virtualization and the blue section refers to the typical resource types available within cloud computing that we just discussed. That has now brought me to the end of this lecture, coming up next, I shall be discussing the different cloud deployment models that are typically referred to in cloud computing.

Cloud Deployment Models

Hello and welcome to this lecture where I shall explain some of the different deployment models used when adopting Cloud technology. Typically, within Cloud computing, there are three different Cloud model types, each offering different levels of:

- management (upravljanje),
- flexibility (prilagodljivost, fleksibilnost),
- security (varnost) and
- resilience (odpornost),

and these are Public, Private and hybrid.



Let's start with the Public Cloud. A Public Cloud model is where a vendor makes available use of shared infrastructure including, but not limited to, compute, storage, database and network resources, that can be provisioned on demand and typically accessed over the internet for Public usage. The consumer will never see the hardware used nor know the exact physical location of their data, but they will be able to specify the geographic region in which it resides to aid with data latency depending on where you end users are located. It makes sense from a design perspective to host your infrastructure as close to the geographical region as your customers or end users are, as this will provide the best overall performance for them. All back and maintenance of the physical location services such as power calling et cetera, along with the physical maintenance of hosts such as hardware failures, will be maintained by the Cloud vendor and seemingly invisible to the end user. As a general rule, you can access your services on the Public Cloud from anywhere as long as you have an internet connection.

A Private Cloud is different to a Public Cloud in that the infrastructure is Privately hosted, managed, and owned by the individual company using it, giving greater and more direct control of it's data. Enterprises who wish to keep a tighter grasp of security control may adopt this architecture. As a result, the hardware is usually held on premise. How this differs from a typical on-premise server farm approach, is that the same Cloud principals are applied to the design such as the use of virtualization, creating a pool of shared computer storage and network resources, making use of scalability and on-demand provision. With this approach, more capital expenditure is requited to acquire the host and the data center that they will physically reside in. Not only this, additional resource will be needed for the day to day operations and maintenance of this equipment. And so your daily operational cost will also increase compared to that of a Public Cloud model.

As you may have already guessed, a hybrid Cloud is a model that makes use of both Public and Private Clouds. This model may be used for seasonal burst traffic or for Disaster Recovery. A hybrid model is established when a network link is configured between the Private Cloud to services within the Public Cloud, essentially extending the logical internal network of the Priv ate Cloud.

This makes the benefits given from both the Public and Private models and allows you to architect your services in the most appropriate model. However, be aware that they also contain the same negatives from both solutions too. Hybrid Clouds are normally short-term configurations, maybe for test and dev purposes, and can often be a transitional state for enterprises before moving a service to reside purely in the Public Cloud.

	Public	Private	Hybrid
Security	Adopts a shared responsibility model between the Vendor and the consumer. The Vendor will manage physical security of the datacenter both internally and externally adhering to a range of Security audit controls. The consumer is responsible for the Security within the Cloud utilizing available tools, services and configurations	Security becomes the sole responsibility of the enterprise adopting the Private cloud. Strict governance can be applied at every point throughout the infrastructure stack to the standards that are desired.	Combination of Public and Private allowing enterprises to have stricter controls over security for data hosted in their private cloud, while taking advantage of the benefits of the public cloud.
Data Location	Consumers specify a geographic location to deploy their services and data, but never know the exact physical location of where that data physically resides	Organizations have full knowledge on the location of their data as it is self managed and held on-premise	Combination of Public and Private
Capital Expenditure	None required - All hardware and maintenance of resources are provisioned by the vendor	High - As hardware is housed and maintained on-premise equipment needs to be procured	Combination of Public and Private
Operational Expenditure	Variable - Simply pay for the resources that you use only when you are using them	High - Additional resources required to maintain and run the hardware, along with the ongoing costs of space, power and cooling within the Data Centre	Combination of Public and Private
Tenancy	Resources can be shared between multiple tenants, in addition to dedicated resources for dedicated tenants	Resources are privately shared for single tenant use	Combination of Public and Private

This table highlights some of the differences between the Cloud types. So feel free to pause the video just to examine the contents of this table just so you really understand the differences. That brings me to the end of this lecture covering the three main deployment types of Cloud technology. For the remainder of this course, I'll be primarily focusing on Public Cloud deployments (namestitev; postavitev). Coming up in the next lecture, I'll be looking at some of the key Cloud concepts to be aware of.

Key Cloud Concepts (characteristics)

Hello and welcome to this lecture aimed at providing you with an understanding of some of the concepts which make cloud technology so important. Cloud computing has a number of key characteristics that allow it to be the powerful service that is it today, and it's a good idea to have an understanding and awareness of these and what they offer.

On-demand resourcing (zagotavljanje virov na zahtevo). This essentially means that when you want to provision a resource within the cloud, it's almost immediately available to you to allocate where and when you need it. No more waiting around for hardware to be ordered, installed, cabled, and configured before using it.

Scalability (skalabilnost; razširljivost). Cloud computing offers you the ability to rapidly scale your environment's resources both up and down and in and out, depending on your requirements and demands of your application and services. When scaling **up and down**, you effectively alter the power and performance of an instance, perhaps using one with a greater CPU or memory power. When scaling **in and out**, you are simply adding or removing the number of instances you're using to your fleet of compute resources. This offers a significant advantage compared to on-premise solutions, from a cost-perspective alone.

Economy of scale (ekonomija obsega). Due to the huge scale of resources public cloud offerings provide, which are optimized and shared between different organizations, thanks to virtualization technology, you as the end user benefit from exceptionally low resource costs compared to traditional hosting.

Flexibility and elasticity (prilagodljivost in elastičnost). Cloud computing offers huge flexibility and elasticity to your design approach. You can choose to have as many or as few resources as you require. You decide how much and how long you want it for, and at what scale. The amount of choice you have allows you to fully customize exactly how you want and need your environment, using only the resources required.

Growth. Cloud computing offers your organization the ability to grow using a wide range of resources and services. Couple with the on-demand element, and your growth constraints are significantly reduced compared to a classic on-premises environment. This growth also includes the ability to reach global customers with ease, by provisioning resources across the cloud vendor's global network.

Utility-based metering. With many cloud services, you only pay for what you use. What do I mean by this? If you only have an instance running for two hours, and then shut it down, then you only pay for two hours worth of compute resources, and that's it. Think of it like this. In your house, you only pay for your electricity when you use it, and to help keep costs down, you turn off the lights when you're not using them. So it's the same billing process for many resources and services. You only pay for resources when you are using them.

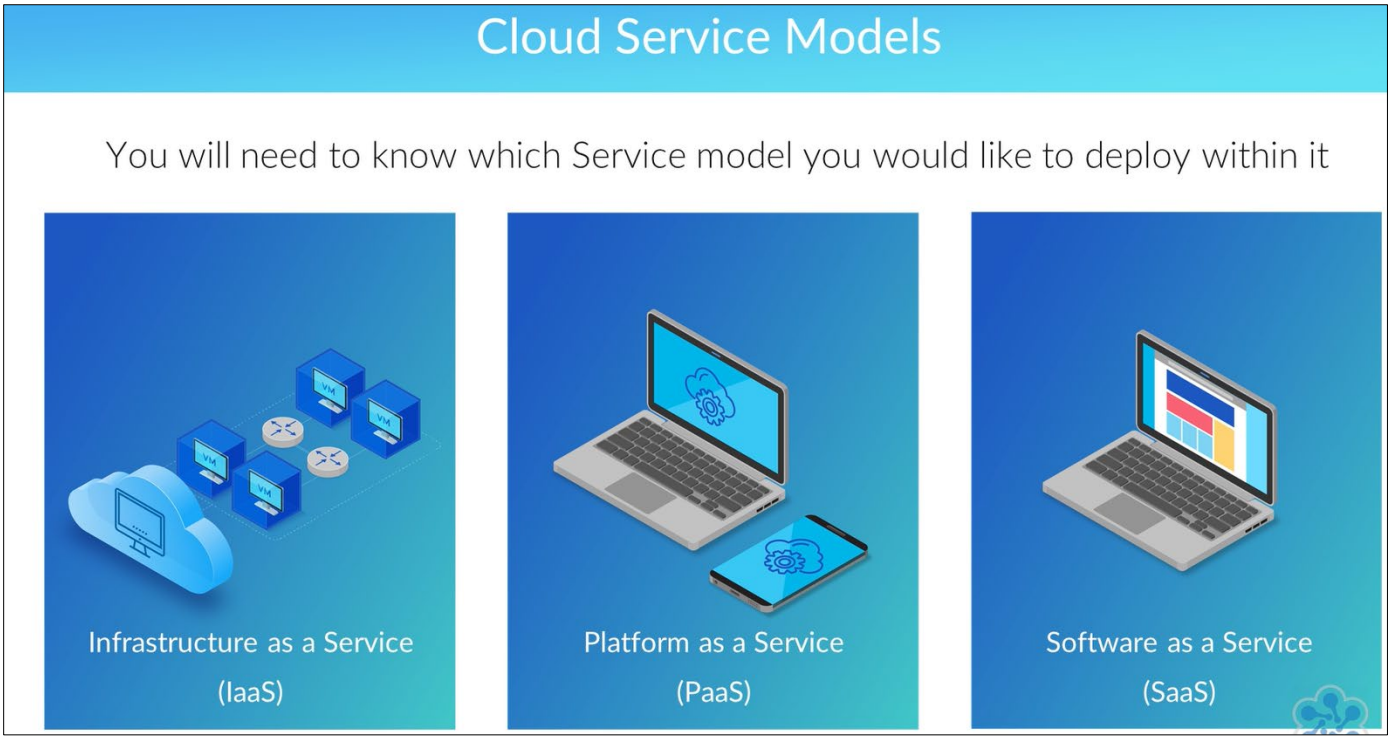
Shared infrastructure. As discussed previously, during the virtualization section, hosts within the cloud are virtualized. As a result, multiple tenants can be running instances on the same piece of hardware. This significantly reduces the amount of physical hardware required, which in turn reduces the amount of power, cooling, and space required in the data center. And in turn, helps with the economy of scale, all resulting in cheaper costs to you as the customer.

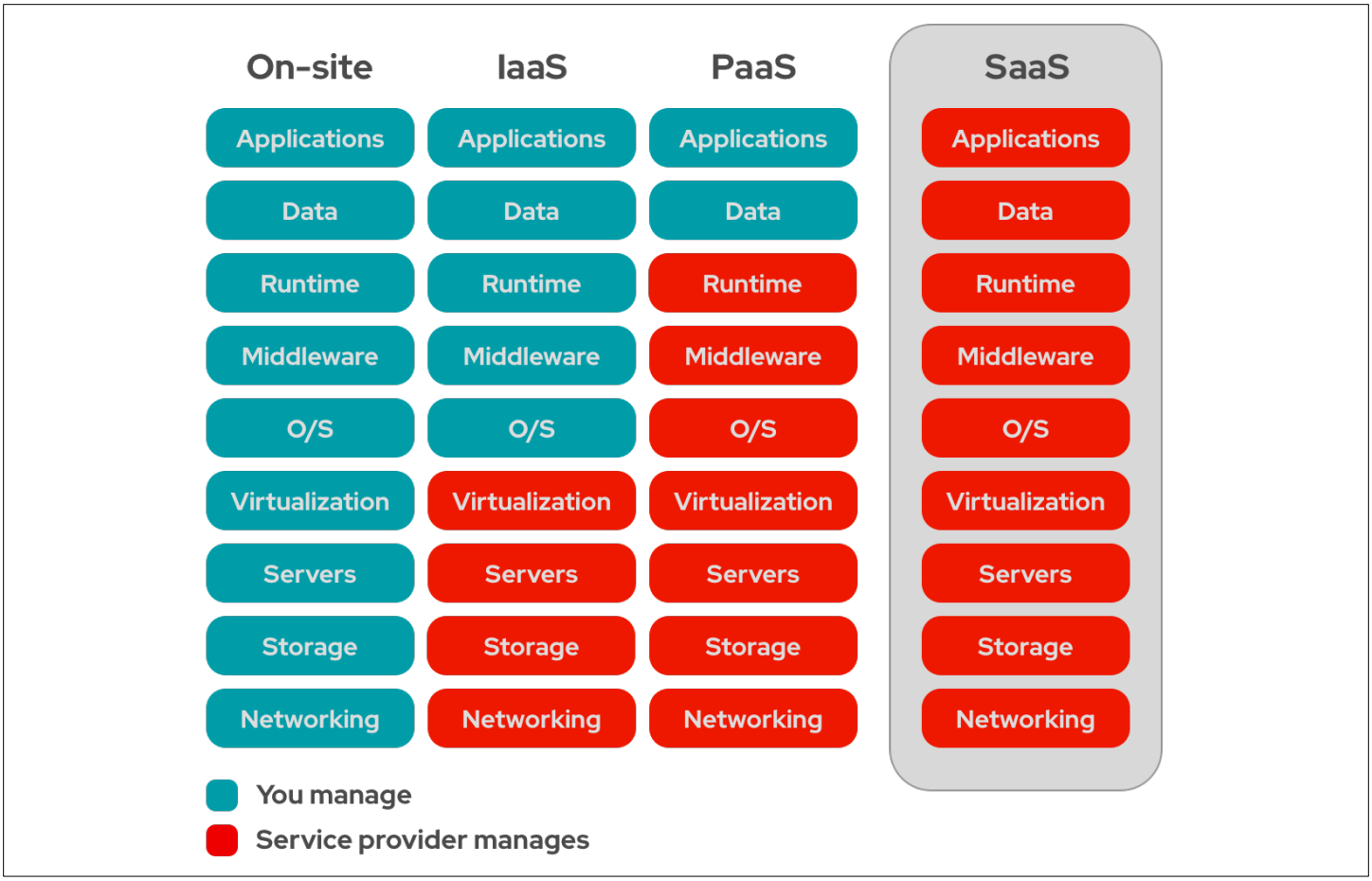
Highly available. By design, many of the core services within the public cloud and its underlying infrastructure are replicated across different geographic zones and regions. Having data copied to multiple different places automatically, helps you to ensure the **durability** and **availability** of your data and services, without even having to configure and architect for this resilience. It's provided by the vendor as a part of their service.

Security. This is one of the most discussed topics within cloud computing, and many enterprises still have concerns over how secure it is. However, public cloud vendors such as AWS and Microsoft Azure are **considered to be more secure than your own data center**. This is down to the fact that they have to adhere to global compliance programs across multiple industries and by applying the **shared responsibility model**. The vendor will operate to an exceptionally high standard of security for the underlying infrastructure of the cloud, and it's down to you, the end user, to then architect security in the cloud, using the tools, services and applications available. For more information on the shared responsibility model for Amazon Web Services, please take a look at our existing blog here. These are just some of the key characteristics of cloud computing, and you can see how differently it operates from your traditional on-premise data center deployments that you may be using today. That now brings me to the end of this lecture. Next I'll be looking at the different cloud service models available.

Cloud Service Models

Hello and welcome to this lecture, whilst you'll be looking at some of the different cloud service models that you will likely encounter as you become more familiar with cloud computing. So now you have an idea of the different cloud types that there are, public, private, and hybrid. You will need to know which service model you would like to deploy within it.





Vir: https://www.redhat.com/en/topics/cloud-computing/what-is-saas

There are many different **service models** available, and more being defined all the time, although three of the most common are that of infrastructure as a service, platform as a service, and software as a service. Each service offering provides a **different level** of **manageability** and **customization** over your solution. So let's look at the **lowest level of customization first**, that being

Software as a service. You would have used many examples of software as a service application, perhaps without even realizing it. Software as a service allows for the delivery of an application that can be widely distributed and accessed. An example of this would be Google's email service, **Gmail**. This email-based application is fully managed by Google and is accessed over the internet, and there are no requirements to install any software on your local device to be able to use it. They are usually simple in their design, focusing on the ease of use to appeal to the wider audience. From the user perspective, this offers the least amount of customization to the application itself.

Platform as a service. This service offering gives a greater level of management and control to you, as you have access to an application framework that sits on top of the operating system and up. The underlying architecture, the host hardware, network components, and operating system are typically managed by the vendor and taken care of from a maintenance and support perspective, which makes this a **great deployment service for developers**. Developers are then free to focus and concentrate on developing great new apps sitting on top of the platform. (Examples: https://www.webhostingsecretrevealed.net/blog/web-business-ideas/paas-examples/)

Infrastructure as a service. This provides the **highest level of customization and management**. This service allows you to architect your own portion of the cloud by configuring a virtual network, which is segmented from other networks, allowing you to deploy any resources you require. In addition to this, you have the ability

to configure instances from the operating system and up, including the type of operating system you install. This service offers the highest level of customization.

However, the underlying host is still managed by the vendor for maintenance and security purposes. There are a number of other service models, such as disaster recovery as a service, communications as a service, and monitoring as a service. For the purpose of this course, we do not need to delve into these. However, I wanted you to be aware that there are more of these services that are making their way into the industry. For now, though, as this is an introduction, you simply need to focus and be aware of software as a service, platform as a service, and infrastructure as a service, as these are the most common within the industry. That has brought me to the end of this lecture. Coming up next, I shall be explaining some of the common use cases of cloud computing.

Common Use Cases of Cloud Computing

Hello and welcome to this lecture. Looking at some of the use cases of cloud computing, and how enterprises have adopted this technology. Following the cloud concept section discussed previously you may already be thinking of some of your own uses that you could use a public cloud for. However, in this lecture I just want to cover some of the common use cases of **why organizations implement cloud computing**. I want to start by touching one of the bigger use cases where people migrate production services from their existing on-premise solutions into the cloud. We have discussed the benefits of the cloud, and so with all those in mind many businesses are choosing to do just that.

Migrate their existing production services to the public cloud. Some companies even have all of their infrastructure within the cloud.

Traffic Bursting. As an example you may experience times within the year perhaps for predicted seasonal circumstances where your infrastructure takes a heavier load impact than that of other times of the year. Perhaps you are in the **retail business**, and over the Christmas holiday period **demand increases on your infrastructure significantly**. In a classic data center environment you would need to provision your compute storage database, and network capacity to reflect this additional traffic, and have it take up additional space, power, and cooling all of the time. This is not an effective method of scaling not only this, but there will be additional costs for this extra infrastructure to **obtain**, **maintain** and **operate**, and you may **only use it for a couple of months of the year.** A far better method of handling this peak traffic road will be to look at cloud computing. The public cloud can be used to scale your networking resources to manage, and handle this additional traffic over your peak season. When the traffic has reduced you can then terminate your infrastructure within this cloud and stop paying for it. Remember **you only pay for what you use when you use it.**

Backup and Disaster Recovery. Due to the public clouds built-in resiliency, and durability, this makes way for a great solution for your backup requirements. To a degree you have access to unlimited storage space with built-in data management lifecycle policies allowing you to make use of even cheaper storage. For example, using Amazon Web Services S3 service, you can implement a policy to archive any data that is over 30 days old to

another service called AWS Glacier which is a cold storage service with an even lower storage cost. The data is still available to you for as long as you have access to the Internet. These storage services are often replicated by the vendor to ensure its durability. Couple that with a very low cost of storage, and you can see why more and more enterprises are adopting cloud computing for this very reason.

Web Hosting. Many organizations choose to host their web services on the cloud due to its ability to load balance across multiple instances as well as scale up and down quickly and automatically as traffic increases and decreases with demand. The ability to provision, and implement **automatic scaling** simplifies the whole process and takes out much of the administrative input, and maintenance required. Not only can your web application, and database service be enhanced by design, but they can also make use of other services such as a content delivery network a CDN, and Domain Name Services DNS.

Remember earlier when we were talking about selecting a geographic region for your instance depending on where end-users are. Well, what if you had end-users all over the world? A CDN is a set of systems which redirects traffic to the closest caching server which can deliver the content much faster. As a result, a CDN can **reduce the latency** of a website for users across the globe if there are sufficient caching servers in place. DNS services can help to manage demand on your web servers by redirecting any requests to a load balancer first. This load balancer can then evenly distribute the requests to multiple web instances that you may have, therefore, reducing demand on a particular web server.

Test and Development Environments. Similarly to our first point of traffic bursting, you may not have the capacity to hosting lots of servers, and storage in your data center for tests, and development purposes and from a financial perspective this would be a huge expense. Using the public cloud allows you to spin up instances as and when you need them, and then shut them down when finished. This also allows you to provision the size, and capacity of your compute resources say, for example, if you need a high-end powerful instance for your testing for an hour you can have it.

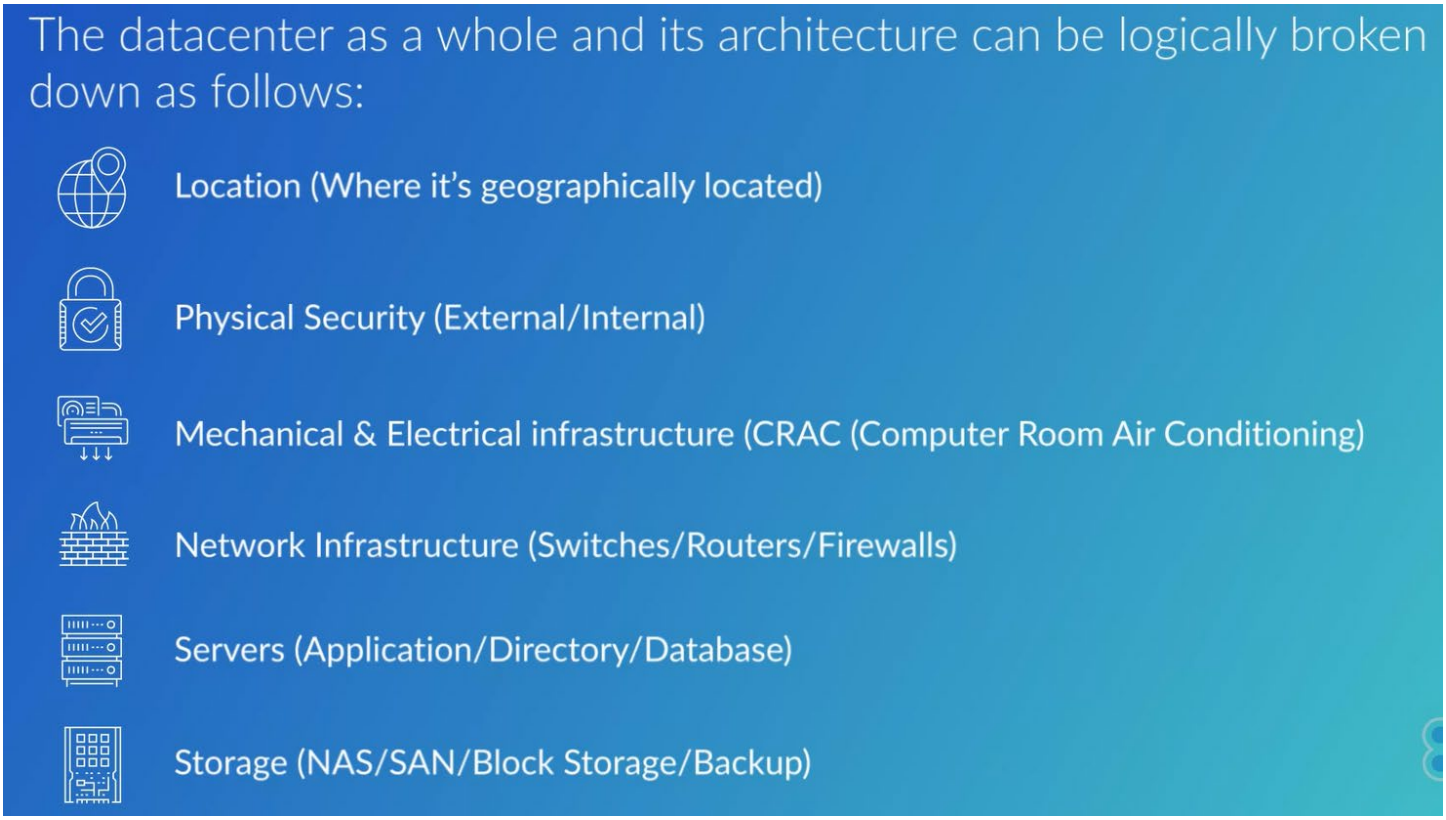
It would not be financially viable for you to have this wide range of compute resource within your own data center.

Proof of Concept (preizkus koncepta). The cloud easily allows you to implement a proof of concept design and ideas to help bring them to life at a fraction of the cost for the reasons that I've already covered in the previous points. This includes hosting costs and only paying for what you use. The results of your proof-of-concept can help you to build a successful business case when presenting to senior management.

Big Data and Data Manipulation. The cloud also makes it easier, and cheaper to manage big data. Maintaining and implementing compute resources to handle huge data sets can be expensive and complicated. By using cloud computing resources you have the ability to use only the resources you need to analyze data when you need it. Some public cloud vendors offer specialized managed Big Data services. Which gives you a managed resource infrastructure, and framework to run your workloads on, in addition to allowing for scalability, scheduling, monitoring et cetera. Having some of these elements managed by the vendor allows you to focus on the data, and processing and not worry about the maintenance, or the underlying architecture. There are many many more use cases of services within the cloud computing space, and you'll more than likely have a few ideas, or requirements of your own. Whatever you choose to do you will have the benefits of the concepts discussed earlier at your disposal. That now brings me to the end of this lecture. Coming up next I shall look at the comparisons between Data Center Architecture and that of the cloud.

How Data Centre architecture is reflected within Cloud

Hello and welcome to this lecture. Where I want to explain how certain technology that we are used to on premises such as routers and switches are reflected within the public cloud. You may be new to cloud computing, but you may have the experience of architecture from a classic on premise solution within a datacenter. This lecture assumes you have the knowledge of what an on premise datacenter is and how they operate. And during this lecture will make comparisons between those datacenters and how cloud computing operates. The datacenter as a whole and its architecture can be logically broken down as follows.



Location. where it's geographically located. Physical Security, both External and Internal. Mechanical and Electrical Infrastructure such as Computer Room Air conditioning units, generators, UPS and fire suppression. Network Infrastructure. Including Switches, Routers and Firewalls. Servers. Such as Application Director and Database servers. Storage. For example, your NAS, SAN Block Storage and Backup.

Let's Look at each of these in turn and discuss starting with Location. Depending on the size of your enterprise, you may only have one office and as a result, you house all of your infrastructure on site. However, many of you probably have multiple datacenters to help with resilience and availability. Public cloud providers do the same. They will have regions all over the globe. And with each of these regions, they will have at least two datacenters. These datacenters will be in a different geographic location within that region.

But close enough to provide high speed interconnectivity between them for data transfer in addition to assisting with high availability and resilience.

Physical Security. As we have already discovered, the Public cloud is operated, managed and maintained by the vendor. As a result, the end user has no access to the physical datacenter where the resources are located. It is the vendors responsibility to ensure it is implementing and achieving the correct certification and governance regarding security. Public cloud vendors adhere to the most stringent of security controls, and for audit purposes, you can access our accreditation and compliance certifications online if required to do so. I've

included links to two of the leading public cloud vendors, where you can find more information about their security and compliance:

https://aws.amazon.com/compliance/

https://www.microsoft.com/en-us/trust-center/compliance/compliance-overview

Mechanical and Electrical. Mechanical and Electrical Infrastructure such as generators, UPS systems, Computer Room Air conditioning units for cooling, Fire suppression et cetera is situated at the datacenter itself. As a result, the same rule applies. It is the vendors responsibility to ensure they are implementing the correct capacity, resiliency and testing to ensure availability and uptime of their infrastructure. So again, this burden is removed from the end user. Where roles and responsibilities begin to change between vendor and the customer is from this point onwards. Network Infrastructure. Network Infrastructure can be quite an extensive list of equipment. However, I'm going to cover some of the more common components and how they sit within the public cloud. Networking operates at software level, there are no options to install a switch or router. How this is achieved is beyond the scope of this course.

But what is important is this, to a degree, you are able to implement controls, services and configurations to simulate the same effects within your infrastructure that these devices provide. There are options to create a virtual network, which can be segmented into different IP address ranges allowing you to deploy compute storage and other network resources as you require. These virtual networks are usually simple to set up requiring just a few small details before it's created. Within AWS, these are classed as Virtual Private Clouds VPCs whereas Microsoft Azure refers to these networks as an Azure Virtual Network VNet. From here you can create different network segments both public facing and private. Routing and Access Control List for security can also be configured dynamically for enhanced control and once you have created your logical virtual network, and of course different geographic regions If desired you can then provision your instances within these different subnets. So networking components such as switches, routers and even firewalls have been replaced with virtual networks and their configurable components.

The backend element of how these services work is again maintained and managed by the vendor. But architecting how your virtual network is created is your responsibility and your responsibility to ensure it secure and not let vulnerable to attacks.

Servers. Depending on your vendor, servers are typically referred to as instances or virtual machines VMs. But as there are different servers within the typical datacenter, vendors provide different services to affect them in the cloud. For example, providers offer servers that are specific to hosting databases, and others focused on heavy processing power that is needed to process big data. This is part of the benefit of the cloud. The vendors are capable of replicating the functions that you use with your in house datacenter but with a lot more advantages.

Storage. Storage is fantastic within the Cloud as it's often regarded as unlimited, hugely scalable and highly durable. As with compute power, there are different storage services depending on what sort of data you're using and where you intend to use it. Within your datacenter environment, you may have access to a Storage Area Network a SAN.

Public cloud providers can also provide block level storage too by their services. AWS offers the Elastic Block Store EBS service. Which offers persistent block level storage, and can we detach from one instance and re-attached to another instance again all configured and deployed within a few clicks. There are other great storage services within the public cloud that cater for file level storage and object storage too. And between the leading public cloud vendors, AWS and Azure, they host a myriad of storage services to cater to different solutions. For more information on these storage services, please see the following links.

https://aws.amazon.com/products/storage/

https://azure.microsoft.com/en-us/services/storage/

Now I've only skimmed the surface of the available services and offer within the public cloud. And there are many other great services that offer amazing solutions to requirements. Many organizations have within those ideas date. However, I just wanted to point out how your current datacenter solutions and elements easily translate into the public count environment. Was at the same time carrying significant advantages and benefits as discussed earlier in this course. In the next and final lecture, I shall be providing a high-level summary of the key points taken throughout this course.

Summary

Hello and welcome to the final lecture within this course, which will recap some of the main points at a high level that I raised throughout the previous lectures. And within these lectures, we learned that

- cloud computing is a remote virtual pool of on-demand shared resources offering compute, storage, database, and network services that can be rapidly deployed at scale.
- Cloud computing is based on virtualized technology.
- Typical cloud foundational resources fall under the categories of compute, storage, database, and network.
- And there are three main cloud deployment models, these being public, private, and hybrid.
- The main concepts of cloud computing are on-demand, scalability, economies of scale, flexibility, growth, utility-based metering, shared infrastructure, highly available, and security as shared responsibility.
- And there are three main cloud service models, infrastructure as a service, platform as a service, and software as a service.

And there are a number of different use cases for cloud computing, and among them are migration of production services, traffic bursting, backup and disaster recovery, web hosting, test and development environments, proof of concepts, and big data and data manipulation. That has now brought me to the end of this lecture and to the end of this course. I hope you now have a greater understanding of what cloud computing is and how it can be used, deployed, and the benefits it offers offer a typical data center deployment. If you have any feedback on this course, positive or negative, please contact us by sending an email to support@cloudacademy.com. Your feedback is greatly appreciated. Thank you for your time and good luck with your continued learning of cloud computing. Thank you.